Answers and talk notes

Represent Polygons on the Coordinate Plane

What your student is learning

Tonight a grid unit stands for real feet — so we count units, convert to feet, and solve a real design problem.

Answers

Ask your student to explain before you look.

1. **B.** 20 feet. 4 × 5 = 20 feet.
2. **C.** 3 units. 30 ÷ 10 = 3 units.
3. **A.** 6 units. Both y-values are positive, so subtract: 7 – 1 = 6 units.
4. **D.** 120 feet. The side is 2 + 4 = 6 units, and 6 × 20 = 120 feet.
5. **D.** 4,000 square feet. The sides are 80 feet and 50 feet, so the area is 80 × 50 = 4,000 square feet.

Talk about it

How many grid units is the side? What does each unit stand for in real feet?

Home mission

Choose a hallway or a counter. Let one step stand for 2 feet. Ask your student: if the hallway is 7 steps, how many feet is that? Then flip it: a 10-foot wall is how many steps?

When they are stuck, ask

- What does one grid unit stand for here?
- Is this number in grid units or in feet?
- Does this distance cross zero? Then what do we do?

Try not to say

- Grid units and feet are the same thing.
- Just subtract the coordinates, the signs will work out.
- Convert at the very end, it does not matter when.